

Matrices



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Definition and Transpose



Introduction: Definition and Transpose

- ▶ Matrices are fundamental objects in linear algebra, extending the concept of vectors to rectangular arrays of numbers
- ▶ We will see how to define them and introduce the transpose operation
- ▶ We also fix the notation for rows, columns and minor matrices, which will be used for determinants and rank

Definition 1: Matrix

A **matrix** is a rectangular table of real numbers with m rows and n columns.

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, we write:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Terminology:

- ▶ The integers m and n are the **dimensions** of the matrix: m is the number of rows and n is the number of columns.
- ▶ The quantity a_{ij} denotes the **entry** of $\mathbf{A}$ in row i and column j , for $i \in \{1, 2, \dots, m\}$ and $j \in \{1, 2, \dots, n\}$.

Example

The matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$$

is a matrix with $m = 2$ rows and $n = 3$ columns, where $a_{11} = 1$, $a_{12} = 2$, $a_{13} = 3$, $a_{21} = 4$, $a_{22} = 5$, and $a_{23} = 6$.

Definition 2: Transpose of a matrix

Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **transpose** of $\mathbf{A}$, denoted by $\mathbf{A}'$, is the $n \times m$ matrix obtained by interchanging rows and columns:

$$\mathbf{A}' = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{n \times m}. \quad (1)$$

The entry in position (i, j) of the transpose matrix $\mathbf{A}'$ is:

$$a'_{ij} = a_{ji}, \quad \forall i \in \{1, 2, \dots, n\}, \quad \forall j \in \{1, 2, \dots, m\}.$$

Example

Given the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3},$$

its transpose is

$$\mathbf{A}' = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \in \mathbb{R}^{3 \times 2}.$$

Rows and Columns

Notation:

- ▶ The i -th row of a matrix $\mathbf{A}$ is denoted by

$$R_i(\mathbf{A}) \quad \text{or} \quad \mathbf{A}_i$$

- ▶ The j -th column of a matrix $\mathbf{A}$ is denoted by

$$C_j(\mathbf{A}) \quad \text{or} \quad \mathbf{A}_j$$

- ▶ In both cases, we can simply use R_i or C_j when the matrix is clear from the context.

Minor Matrices

Let $\mathbf{A} \in \mathbb{R}^{n \times m}$ be a matrix and let $i \in \{1, 2, \dots, n\}$, $j \in \{1, 2, \dots, m\}$.

We denote by $\mathbf{A}_{ij}$ the $(n - 1) \times (m - 1)$ matrix obtained from $\mathbf{A}$ by deleting row i and column j .

The matrix $\mathbf{A}_{ij}$ is called the **minor matrix** associated with the entry a_{ij} .

Example

Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 & 5 \\ 3 & 4 & 7 & -1 \\ 2 & 0 & -3 & 6 \\ -4 & 1 & 2 & 8 \end{pmatrix} \in \mathbb{R}^{4 \times 4}.$$

For $i = 2$ and $j = 3$, the minor matrix $\mathbf{A}_{23} \in \mathbb{R}^{3 \times 3}$ is obtained by deleting row 2 and column 3:

$$\mathbf{A}_{23} = \begin{pmatrix} 1 & -2 & 5 \\ 2 & 0 & 6 \\ -4 & 1 & 8 \end{pmatrix}.$$

Summary

Matrices:

- ▶ A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a table of real numbers with m rows and n columns
- ▶ a_{ij} is the entry in row i and column j ; $R_i(\mathbf{A})$ and $C_j(\mathbf{A})$ denote its i -th row and j -th column

Transpose and minor matrices:

- ▶ The transpose $\mathbf{A}' \in \mathbb{R}^{n \times m}$ interchanges rows and columns:
 $a'_{ij} = a_{ji}$
- ▶ The minor matrix $\mathbf{A}_{ij}$ is obtained by deleting row i and column j

Sum and Scalar Multiplication



Introduction: Sum and Scalar Multiplication

- ▶ We define sum and scalar multiplication for matrices, similar to vectors but operating on tables of numbers
- ▶ Both operations act **entry by entry**
- ▶ They satisfy the same algebraic properties as the corresponding operations on vectors

Sum of matrices

Given two matrices

$$\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$$

their sum is the matrix

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix} \quad (2)$$

Scalar multiplication

Given a scalar $\lambda \in \mathbb{R}$ and a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the **scalar multiplication** of $\mathbf{A}$ by λ is the matrix

$$\lambda \mathbf{A} = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ \lambda a_{21} & \lambda a_{22} & \cdots & \lambda a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda a_{m1} & \lambda a_{m2} & \cdots & \lambda a_{mn} \end{pmatrix} \quad (3)$$

Example

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 4 & 5 \\ -1 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 1+4 & -2+5 \\ 3+(-1) & 0+2 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 2 & 2 \end{pmatrix},$$

and, for $\lambda = -2$,

$$\lambda \mathbf{A} = -2 \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ -6 & 0 \end{pmatrix}.$$

Properties of matrix sum and scalar multiplication (1/2)

The main properties of matrix sum and scalar multiplication are:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \quad \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n} \quad (4)$$

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}), \quad \forall \mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{m \times n} \quad (5)$$

$$\mathbf{A} + \mathbf{0} = \mathbf{A}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (6)$$

$$\mathbf{A} + (-\mathbf{A}) = \mathbf{0}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (7)$$

Properties of matrix sum and scalar multiplication (2/2)

$$\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}, \quad \forall \lambda \in \mathbb{R}, \forall \mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n} \quad (8)$$

$$(\lambda + \mu)\mathbf{A} = \lambda\mathbf{A} + \mu\mathbf{A}, \quad \forall \lambda, \mu \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (9)$$

$$\lambda(\mu\mathbf{A}) = (\lambda\mu)\mathbf{A}, \quad \forall \lambda, \mu \in \mathbb{R}, \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (10)$$

$$1 \cdot \mathbf{A} = \mathbf{A} \quad \text{and} \quad 0 \cdot \mathbf{A} = \mathbf{0}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n}. \quad (11)$$

Summary

Operations:

- ▶ **Sum:** $\mathbf{A} + \mathbf{B}$ (same dimensions) is obtained by adding the entries in the same position
- ▶ **Scalar multiplication:** $\lambda \mathbf{A}$ is obtained by multiplying every entry by λ

Properties:

- ▶ The sum is commutative and associative, 0 is the neutral element and $-\mathbf{A}$ is the opposite of $\mathbf{A}$
- ▶ Scalar multiplication is distributive, $\lambda(\mu \mathbf{A}) = (\lambda\mu) \mathbf{A}$, $1 \cdot \mathbf{A} = \mathbf{A}$ and $0 \cdot \mathbf{A} = 0$

Matrix Product

Introduction: Matrix Product

- ▶ The matrix product is a fundamental operation that combines two matrices
- ▶ Unlike addition, it requires specific dimension compatibility: the number of columns of the first matrix must equal the number of rows of the second one
- ▶ Each entry of the product is a **scalar product** between a row and a column
- ▶ We will see that, in general, the product is **not commutative**

Matrix product

Given two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$, their multiplication is the matrix

$$\mathbf{AB} = \begin{pmatrix} \sum_{j=1}^n a_{1j}b_{j1} & \sum_{j=1}^n a_{1j}b_{j2} & \cdots & \sum_{j=1}^n a_{1j}b_{jp} \\ \sum_{j=1}^n a_{2j}b_{j1} & \sum_{j=1}^n a_{2j}b_{j2} & \cdots & \sum_{j=1}^n a_{2j}b_{jp} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{j=1}^n a_{mj}b_{j1} & \sum_{j=1}^n a_{mj}b_{j2} & \cdots & \sum_{j=1}^n a_{mj}b_{jp} \end{pmatrix} \in \mathbb{R}^{m \times p} \quad (12)$$

where the entry in position (i, k) , i.e., $[\mathbf{AB}]_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$, is the scalar product of the i -th row of $\mathbf{A}$ and the k -th column of $\mathbf{B}$.

Example (1/2)

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

The product $\mathbf{AB}$ is:

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} \\ &= \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix} \end{aligned}$$

Example (2/2)

The product $\mathbf{BA}$ is:

$$\begin{aligned}\mathbf{BA} &= \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 5 \cdot 1 + 6 \cdot 3 & 5 \cdot 2 + 6 \cdot 4 \\ 7 \cdot 1 + 8 \cdot 3 & 7 \cdot 2 + 8 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 23 & 34 \\ 31 & 46 \end{pmatrix}\end{aligned}$$

Note that $\mathbf{AB} \neq \mathbf{BA}$.

Example (1/2)

Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 3} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 2}$$

The product $\mathbf{AB}$ is:

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 3 & 1 \cdot 1 + 2 \cdot 0 + (-1) \cdot 5 \\ 0 \cdot 2 + 3 \cdot (-1) + 4 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 + 4 \cdot 5 \end{pmatrix} \\ &= \begin{pmatrix} -3 & -4 \\ 9 & 20 \end{pmatrix} \end{aligned}$$

Example (2/2)

The product $\mathbf{BA}$ is:

$$\begin{aligned}\mathbf{BA} &= \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \cdot 1 + 1 \cdot 0 & 2 \cdot 2 + 1 \cdot 3 & 2 \cdot (-1) + 1 \cdot 4 \\ -1 \cdot 1 + 0 \cdot 0 & -1 \cdot 2 + 0 \cdot 3 & -1 \cdot (-1) + 0 \cdot 4 \\ 3 \cdot 1 + 5 \cdot 0 & 3 \cdot 2 + 5 \cdot 3 & 3 \cdot (-1) + 5 \cdot 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 7 & 2 \\ -1 & -2 & 1 \\ 3 & 21 & 17 \end{pmatrix}\end{aligned}$$

Again, $\mathbf{AB} \neq \mathbf{BA}$.

Properties of matrix product (1/2)

The main properties of the matrix product (when dimensions are compatible) are:

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) \quad (13)$$

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC} \quad (14)$$

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC} \quad (15)$$

$$\lambda(\mathbf{AB}) = (\lambda\mathbf{A})\mathbf{B} = \mathbf{A}(\lambda\mathbf{B}) \quad (16)$$

Properties of matrix product (2/2)

$$\mathbf{I}_m \mathbf{A} = \mathbf{A} \quad \text{and} \quad \mathbf{A} \mathbf{I}_n = \mathbf{A}, \quad \forall \mathbf{A} \in \mathbb{R}^{m \times n} \quad (17)$$

In general, the matrix product is **not commutative**:

$$\mathbf{AB} \neq \mathbf{BA} \quad \text{in general.} \quad (18)$$

Summary

Matrix product:

- ▶ $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$ give $\mathbf{AB} \in \mathbb{R}^{m \times p}$
- ▶ $[\mathbf{AB}]_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$: scalar product of row i of $\mathbf{A}$ and column k of $\mathbf{B}$

Properties:

- ▶ Associative and distributive; scalars can be moved:
 $\lambda(\mathbf{AB}) = (\lambda\mathbf{A})\mathbf{B} = \mathbf{A}(\lambda\mathbf{B})$
- ▶ $\mathbf{I}_m \mathbf{A} = \mathbf{A} \mathbf{I}_n = \mathbf{A}$
- ▶ In general $\mathbf{AB} \neq \mathbf{BA}$ (not commutative)

Special Matrices

Introduction: Special Matrices

- ▶ Certain matrices with special structures appear frequently in applications and have important properties
- ▶ We introduce square, identity, diagonal, symmetric and triangular matrices
- ▶ We then study permutation matrices (to reorder rows and columns), inverse matrices and semidefinite matrices (fundamental in optimization)

Definition 3: Square matrix

A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ is called a **square matrix** if it has the same number of rows and columns ($m = n$). In this case, $\mathbf{A} \in \mathbb{R}^{n \times n}$ and it has the form:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}. \quad (19)$$

Example

An example of a square matrix of size 3 is:

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \\ 1 & 0 & 7 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Definition 4: Identity matrix

A square matrix $I_n \in \mathbb{R}^{n \times n}$ is called the **identity matrix** if it has ones on the main diagonal and zeros elsewhere. It has the form:

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}. \quad (20)$$

Equivalently, the identity matrix satisfies

$$[I_n]_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases} \quad \forall i, j \in \{1, 2, \dots, n\}.$$

Example

The identity matrix of size 3 is:

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Identity matrix and matrix product

For any matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, we have:

$$\mathbf{I}_m \mathbf{A} = \mathbf{A} \quad \text{and} \quad \mathbf{A} \mathbf{I}_n = \mathbf{A}.$$

Definition 5: Diagonal matrix

A square matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ is called **diagonal** if all its off-diagonal entries are zero. It has the form:

$$\mathbf{D} = \begin{pmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_{nn} \end{pmatrix}. \quad (21)$$

Equivalently, a diagonal matrix satisfies

$$d_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i \neq j.$$

Example

The matrix

$$\mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is diagonal.

Definition 6: Symmetric matrix

A square matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is called *symmetric* if it has the same entries with respect to the main diagonal (i.e., the entry in position (i, j) equals the entry in position (j, i)). It has the form:

$$\mathbf{Q} = \begin{pmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{12} & q_{22} & \cdots & q_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ q_{1n} & q_{2n} & \cdots & q_{nn} \end{pmatrix}. \quad (22)$$

Symmetric matrix

Equivalently, a symmetric matrix satisfies

$$q_{ij} = q_{ji}, \quad \forall i, j \in \{1, 2, \dots, n\}.$$

In matrix form, symmetry can be written as

$$Q' = Q.$$

Definition 7: Upper triangular matrix

A square matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ is called **upper triangular** if all its entries below the main diagonal are zero. It has the form:

$$\mathbf{U} = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}. \quad (23)$$

Equivalently, an upper triangular matrix satisfies

$$u_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i > j.$$

Example

The matrix

$$\mathbf{U} = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 4 & 5 \\ 0 & 0 & -2 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is upper triangular.

Definition 8: Lower triangular matrix

A square matrix $\mathbf{L} \in \mathbb{R}^{n \times n}$ is called **lower triangular** if all its entries above the main diagonal are zero. It has the form:

$$\mathbf{L} = \begin{pmatrix} \ell_{11} & 0 & \cdots & 0 \\ \ell_{21} & \ell_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n1} & \ell_{n2} & \cdots & \ell_{nn} \end{pmatrix}. \quad (24)$$

Equivalently, a lower triangular matrix satisfies

$$\ell_{ij} = 0, \quad \forall i, j \in \{1, 2, \dots, n\} \text{ with } i < j.$$

Example

The matrix

$$\mathbf{L} = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 4 & 5 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}$$

is lower triangular.

Permutation Matrices

Permutation matrices are square matrices obtained by permuting the rows (or equivalently, the columns) of the identity matrix.

They encode permutations of indices and implement row/column reorderings through matrix multiplication.

Permutation matrices (1/3)

Given an ordering (permutation) π of $\{1, \dots, n\}$, we denote by $\mathbf{P}_\pi$ the **permutation matrix** obtained from $\mathbf{I}_n$ by reordering its rows according to π , i.e., placing row $\pi(1)$ first, then row $\pi(2)$, $\dots$, and finally row $\pi(n)$.

Permutation matrices (2/3)

Row reordering by left multiplication.

Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{P}_\pi \in \mathbb{R}^{m \times m}$, the matrix

$$\mathbf{P}_\pi \mathbf{A}$$

consists of the rows of $\mathbf{A}$ reordered according to the permutation π (row $\pi(1)$ becomes the first row, row $\pi(2)$ becomes the second row, etc.).

Permutation matrices (3/3)

Column reordering by right multiplication.

Given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{P}_\pi \in \mathbb{R}^{n \times n}$, the matrix

$$\mathbf{AP}'_\pi$$

consists of the columns of $\mathbf{A}$ reordered according to the permutation π (column $\pi(1)$ becomes the first column, column $\pi(2)$ becomes the second column, etc.).

- Note the transpose: $\mathbf{P}'_\pi$ is obtained from $\mathbf{I}_n$ by reordering its *columns* according to π .

Example (1/2)

Consider $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \in \mathbb{R}^{3 \times 2}$.

Given the desired row ordering $\pi = (3, 1, 2)$ (i.e., row 3 first, then row 1, then row 2), we use the permutation matrix obtained from $\mathbf{I}_3$ by reordering its rows as $(3, 1, 2)$:

$$\mathbf{P}_\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Example (2/2)

Then:

$$\mathbf{P}_\pi \mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 6 \\ 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

The rows of $\mathbf{A}$ have been reordered according to $\pi = (3, 1, 2)$.

Example (1/2)

Consider $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$.

Given the desired column ordering $\pi = (2, 3, 1)$ (i.e., column 2 first, then column 3, then column 1), we use the permutation matrix $\mathbf{P}_\pi$ obtained from $\mathbf{I}_3$ by reordering its rows as $(2, 3, 1)$, and its transpose:

$$\mathbf{P}_\pi = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{P}'_\pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Example (2/2)

Then:

$$\mathbf{A}\mathbf{P}'_{\pi} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{pmatrix}.$$

The columns of $\mathbf{A}$ have been reordered according to $\pi = (2, 3, 1)$.

Definition 9: inverse matrix

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is *invertible* if there exists a matrix $\mathbf{A}^{-1} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{A} \mathbf{A}^{-1} = \mathbf{I} \quad \text{and} \quad \mathbf{A}^{-1} \mathbf{A} = \mathbf{I}.$$

Such a matrix $\mathbf{A}^{-1}$ is called the *inverse* of $\mathbf{A}$.

If it exists, the inverse matrix $\mathbf{A}^{-1}$ is unique.

Definition 10: semidefinite matrices

A square *symmetric* matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is

- ▶ *positive semidefinite* if

$$\mathbf{x}' \mathbf{Q} \mathbf{x} \geq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- ▶ *positive definite* if

$$\mathbf{x}' \mathbf{Q} \mathbf{x} > 0, \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus \{0\}$$

Definition 11: semidefinite matrices (continued)

A square symmetric matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is

- ▶ *negative semidefinite* if

$$\mathbf{x}' \mathbf{Q} \mathbf{x} \leq 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

- ▶ *negative definite* if

$$\mathbf{x}' \mathbf{Q} \mathbf{x} < 0, \quad \forall \mathbf{x} \in \mathbb{R}^n \setminus \{0\}$$

Example (1/2)

The identity matrix

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive semidefinite since:

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 + x_2^2 \geq 0, \quad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

Example (2/2)

The matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

is positive semidefinite since:

$$\begin{aligned} (x_1 \quad x_2) \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 \\ &= x_1^2 + (x_1 - x_2)^2 + x_2^2 \geq 0, \end{aligned}$$

for all $(x_1 \quad x_2)' \in \mathbb{R}^2$.

Summary (1/2)

Special square matrices:

- ▶ **Identity** I_n : ones on the diagonal, zeros elsewhere;
 $I_m \mathbf{A} = \mathbf{A} I_n = \mathbf{A}$
- ▶ **Diagonal**: $d_{ij} = 0$ for $i \neq j$
- ▶ **Symmetric**: $q_{ij} = q_{ji}$, i.e., $\mathbf{Q}' = \mathbf{Q}$
- ▶ **Upper triangular**: $u_{ij} = 0$ for $i > j$; **lower triangular**: $\ell_{ij} = 0$ for $i < j$

Permutation matrices:

- ▶ $\mathbf{P}_\pi$: rows of I_n reordered according to π
- ▶ $\mathbf{P}_\pi \mathbf{A}$ reorders the rows of $\mathbf{A}$; $\mathbf{A} \mathbf{P}'_\pi$ reorders the columns of $\mathbf{A}$

Summary (2/2)

Inverse matrix:

- ▶ $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$; if it exists, the inverse is unique

Semidefinite matrices ($\mathbf{Q}$ symmetric):

- ▶ Positive semidefinite: $\mathbf{x}'\mathbf{Q}\mathbf{x} \geq 0$ for all $\mathbf{x}$; positive definite: $\mathbf{x}'\mathbf{Q}\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$
- ▶ Negative semidefinite: $\mathbf{x}'\mathbf{Q}\mathbf{x} \leq 0$ for all $\mathbf{x}$; negative definite: $\mathbf{x}'\mathbf{Q}\mathbf{x} < 0$ for all $\mathbf{x} \neq 0$

Determinant

Introduction: Determinant

- ▶ The determinant is a scalar value that can be computed from a square matrix and encodes important information about the matrix
- ▶ We give explicit formulas for small matrices, the Laplace expansion for the general case and the simple formula for triangular matrices
- ▶ Finally, we list its main properties: in particular, a matrix is invertible if and only if its determinant is different from zero

Determinant formulas

Determinant of a 1×1 matrix. Let $\mathbf{A} = (a) \in \mathbb{R}^{1 \times 1}$. Then

$$\det(\mathbf{A}) = a. \quad (25)$$

Determinant of a 2×2 matrix. Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2}$. Then

$$\det(\mathbf{A}) = ad - bc. \quad (26)$$

Determinant formulas (continued)

Determinant of a 3×3 matrix (Sarrus rule). Let

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

Then

$$\det(\mathbf{A}) = aei + bfg + cdh - ceg - bdi - afh. \quad (27)$$

Example

Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}.$$

Then

$$\det(\mathbf{A}) = 2 \cdot 4 - (-1) \cdot 3 = 8 + 3 = 11.$$

Example

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 1 & 4 \\ 2 & -1 & 1 \end{pmatrix}.$$

Using Sarrus:

$$\det(\mathbf{A}) = 1 \cdot 1 \cdot 1 + 2 \cdot 4 \cdot 2 + 0 \cdot 3 \cdot (-1) - 0 \cdot 1 \cdot 2 - 2 \cdot 3 \cdot 1 - 1 \cdot 4 \cdot (-1).$$

Hence

$$\det(\mathbf{A}) = 1 + 16 + 0 - 0 - 6 + 4 = 15.$$

Laplace Expansion

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix. The determinant $\det(\mathbf{A})$ can be computed recursively by **Laplace expansion** along any row or column.

For any row $i \in \{1, 2, \dots, n\}$:

$$\det(\mathbf{A}) = \sum_{j=1}^n a_{ij} C_{ij} \quad (28)$$

For any column $j \in \{1, 2, \dots, n\}$:

$$\det(\mathbf{A}) = \sum_{i=1}^n a_{ij} C_{ij} \quad (29)$$

Where C_{ij} is the **cofactor** of a_{ij} : $C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij})$, and $\mathbf{A}_{ij}$ is the minor matrix obtained from $\mathbf{A}$ by deleting row i and column j .

Example (1/2)

Let

$$\mathbf{A} = \begin{pmatrix} -2 & 2 & -3 \\ -1 & 1 & 3 \\ 2 & 0 & -1 \end{pmatrix}.$$

We expand along the second column (it contains a zero):

$$\det(\mathbf{A}) = 2 C_{12} + 1 C_{22} + 0 \cdot C_{32}.$$

Compute the cofactors:

$$\begin{aligned} C_{12} &= (-1)^{1+2} \det \begin{pmatrix} -1 & 3 \\ 2 & -1 \end{pmatrix} \\ &= -((-1)(-1) - 2 \cdot 3) = -(1 - 6) = 5 \end{aligned}$$

Example (2/2)

$$C_{22} = (-1)^{2+2} \det \begin{pmatrix} -2 & -3 \\ 2 & -1 \end{pmatrix} = ((-2)(-1) - 2(-3)) = 2 + 6 = 8.$$

Therefore,

$$\det(\mathbf{A}) = 2 \cdot 5 + 1 \cdot 8 = 18.$$

Determinant of Triangular Matrices

For triangular matrices (upper or lower), the determinant has a particularly simple form: it is the product of the diagonal entries.

If $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**, then

$$\det(\mathbf{U}) = \prod_{i=1}^n u_{ii}. \quad (30)$$

If $\mathbf{L} \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix**, then

$$\det(\mathbf{L}) = \prod_{i=1}^n \ell_{ii}. \quad (31)$$

In particular:

- ▶ For a **diagonal matrix** $\mathbf{D} \in \mathbb{R}^{n \times n}$:

$$\det(\mathbf{D}) = \prod_{i=1}^n d_{ii}.$$

- ▶ For the **identity matrix** $\mathbf{I}_n$:

$$\det(\mathbf{I}_n) = 1.$$

Example

Let

$$\mathbf{U} = \begin{pmatrix} 2 & 3 & -1 \\ 0 & -4 & 5 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then

$$\det(\mathbf{U}) = 2 \cdot (-4) \cdot 3 = -24.$$

Example

Let

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 4 & -2 \end{pmatrix}.$$

Then

$$\det(\mathbf{L}) = 1 \cdot 3 \cdot (-2) = -6.$$

Example

Let

$$\mathbf{D} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Then

$$\det(\mathbf{D}) = 5 \cdot (-2) \cdot 3 = -30.$$

Properties of Determinants (1/2)

The determinant satisfies several important algebraic properties.

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ and $\lambda \in \mathbb{R}$. Then:

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \cdot \det(\mathbf{B}) \quad (32)$$

$$\det(\mathbf{A}') = \det(\mathbf{A}) \quad (33)$$

$$\det(\lambda \mathbf{A}) = \lambda^n \det(\mathbf{A}) \quad (34)$$

Properties of Determinants (2/2)

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}, \quad \text{if } \mathbf{A} \text{ is invertible} \quad (35)$$

$$\det(\mathbf{I}_n) = 1 \quad (36)$$

Remarks on the properties of determinants

- ▶ From $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ and $\det(\mathbf{I}_n) = 1$, if $\mathbf{A}$ is invertible:

$$\det(\mathbf{A}) \cdot \det(\mathbf{A}^{-1}) = \det(\mathbf{AA}^{-1}) = \det(\mathbf{I}_n) = 1,$$

which gives $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$.

- ▶ Property $\det(\lambda\mathbf{A}) = \lambda^n \det(\mathbf{A})$ shows that scalar multiplication affects the determinant by the n -th power of the scalar (not linearly).
- ▶ Property $\det(\mathbf{A}') = \det(\mathbf{A})$ implies that row operations and column operations have symmetric effects on the determinant.

Example (1/2)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 2 \cdot 3 - 1 \cdot 0 = 6, \quad \det(\mathbf{B}) = 1 \cdot 2 - (-1) \cdot 4 = 6.$$

Example (2/2)

The product is:

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 12 & 6 \end{pmatrix}.$$

Then:

$$\det(\mathbf{AB}) = 6 \cdot 6 - 0 \cdot 12 = 36 = \det(\mathbf{A}) \cdot \det(\mathbf{B}).$$

Example

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{A}' = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(\mathbf{A}') = 1 \cdot 4 - 3 \cdot 2 = -2.$$

Therefore, $\det(\mathbf{A}') = \det(\mathbf{A})$.

Example

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \quad \lambda = 2.$$

Then:

$$\lambda \mathbf{A} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 1 \cdot 4 - 2 \cdot 3 = -2, \quad \det(\lambda \mathbf{A}) = 2 \cdot 8 - 4 \cdot 6 = -8.$$

Since $n = 2$, we verify: $\det(\lambda \mathbf{A}) = -8 = 2^2 \cdot (-2) = \lambda^2 \det(\mathbf{A})$.

Example

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

We have:

$$\det(\mathbf{A}) = 2 \cdot 1 - 1 \cdot 1 = 1.$$

The inverse matrix is:

$$\mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}.$$

Then:

$$\det(\mathbf{A}^{-1}) = 1 \cdot 2 - (-1) \cdot (-1) = 1 = \frac{1}{\det(\mathbf{A})}.$$

Invertibility and determinant

A matrix $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

Summary (1/2)

Computing the determinant:

- ▶ 1×1 : $\det(\mathbf{A}) = a$; 2×2 : $\det(\mathbf{A}) = ad - bc$; 3×3 : Sarrus rule
- ▶ Laplace expansion along any row i or column j :
 $\det(\mathbf{A}) = \sum_j a_{ij} C_{ij} = \sum_i a_{ij} C_{ij}$, with $C_{ij} = (-1)^{i+j} \det(\mathbf{A}_{ij})$
- ▶ Expand along a row or column with many zeros
- ▶ Triangular and diagonal matrices: product of the diagonal entries; $\det(\mathbf{I}_n) = 1$

Summary (2/2)

Properties:

- ▶ $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$ and $\det(\mathbf{A}') = \det(\mathbf{A})$
- ▶ $\det(\lambda \mathbf{A}) = \lambda^n \det(\mathbf{A})$
- ▶ $\det(\mathbf{A}^{-1}) = 1 / \det(\mathbf{A})$ if $\mathbf{A}$ is invertible

Invertibility:

- ▶ $\mathbf{A}$ is invertible $\iff \det(\mathbf{A}) \neq 0$

Rank

Introduction: Rank

- ▶ The rank measures the “dimension” of the information contained in a matrix by counting how many rows or columns are truly independent
- ▶ It relies on the notion of linear independence introduced in the chapter on vectors
- ▶ It is crucial for determining whether systems of equations have unique solutions

Definition 12: rank of a matrix

*Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The **rank** of $\mathbf{A}$, denoted by $\text{rank}(\mathbf{A})$, is the maximum number of linearly independent columns of $\mathbf{A}$.*

- ▶ The columns of $\mathbf{A}$ are vectors of $\mathbb{R}^m$, and linear independence is meant in the sense of the chapter on vectors: $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if $\lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$ implies $\lambda_1 = \dots = \lambda_k = 0$.

Rank

- ▶ Equivalently, $\text{rank}(\mathbf{A})$ is the maximum number of linearly independent rows of $\mathbf{A}$.
- ▶ We always have

$$\text{rank}(\mathbf{A}) \leq \min\{m, n\}.$$

- ▶ If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is square, then

$$\text{rank}(\mathbf{A}) = n \iff \det(\mathbf{A}) \neq 0.$$

Example

Let

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix} \in \mathbb{R}^{2 \times 3}.$$

The second row is a multiple of the first one:

$$(2 \quad 4 \quad 6) = 2 (1 \quad 2 \quad 3).$$

Hence the two rows are linearly dependent, so there is only one linearly independent row. Therefore,

$$\text{rank}(\mathbf{B}) = 1.$$

Example (1/2)

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix} \in \mathbb{R}^{3 \times 3}.$$

The third column is the sum of the first two:

$$C_3(\mathbf{A}) = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = C_1(\mathbf{A}) + C_2(\mathbf{A}),$$

hence the three columns are linearly dependent and $\text{rank}(\mathbf{A}) \leq 2$.

Example (2/2)

On the other hand, the first two columns are linearly independent:

$$\lambda_1 C_1(\mathbf{A}) + \lambda_2 C_2(\mathbf{A}) = 0$$

gives $\lambda_1 = 0$ (first entry) and $\lambda_2 = 0$ (third entry). Therefore,

$$\text{rank}(\mathbf{A}) = 2.$$

Consistently, expanding along the first row,

$$\det(\mathbf{A}) = 1 \cdot (1 \cdot 1 - 3 \cdot 1) - 0 + 1 \cdot (2 \cdot 1 - 1 \cdot 0) = -2 + 2 = 0.$$

Summary

Rank:

- ▶ $\text{rank}(\mathbf{A})$ is the maximum number of linearly independent columns of $\mathbf{A}$ (equivalently, of rows)
- ▶ $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$
- ▶ For $\mathbf{A} \in \mathbb{R}^{n \times n}$: $\text{rank}(\mathbf{A}) = n \iff \det(\mathbf{A}) \neq 0$
- ▶ To compute it: look for columns (or rows) that are linear combinations of the others, and check the independence of the remaining ones

Key Takeaways

Summary - Part 1: Basics (1/2)

1. Matrix Definition:

- ▶ A matrix is a rectangular table of real numbers: $\mathbf{A} \in \mathbb{R}^{m \times n}$
- ▶ The transpose $\mathbf{A}'$ interchanges rows and columns
- ▶ Minor matrix $\mathbf{A}_{ij}$: obtained by deleting row i and column j

Summary - Part 1: Basics (2/2)

2. Matrix Operations:

- ▶ **Sum:** Component-wise addition (same dimensions required)
- ▶ **Scalar multiplication:** Multiply each entry by scalar
- ▶ **Product:** $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times p}$ yields $\mathbf{AB} \in \mathbb{R}^{m \times p}$
- ▶ Matrix product is **not commutative**: $\mathbf{AB} \neq \mathbf{BA}$ in general

Summary - Part 2: Special Matrices (1/2)

3. Special Matrix Types:

- ▶ **Square:** $m = n$
- ▶ **Identity:** I_n (ones on diagonal, zeros elsewhere)
- ▶ **Diagonal:** All off-diagonal entries zero
- ▶ **Symmetric:** $Q' = Q$
- ▶ **Triangular:** Upper or lower
- ▶ **Permutation:** Row reordering $P_\pi A$, column reordering AP'_π
- ▶ **Invertible:** $AA^{-1} = I$

Summary - Part 2: Special Matrices (2/2)

4. Semidefinite Matrices:

- ▶ Positive semidefinite: $\mathbf{x}'\mathbf{Q}\mathbf{x} \geq 0$ for all $\mathbf{x}$
- ▶ Positive definite: $\mathbf{x}'\mathbf{Q}\mathbf{x} > 0$ for all $\mathbf{x} \neq 0$

Summary - Part 3: Determinant and Rank (1/2)

5. Determinant:

- ▶ Scalar value for square matrices encoding important information
- ▶ 2×2 : $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
- ▶ 3×3 : Sarrus rule or Laplace expansion
- ▶ Triangular/diagonal: product of diagonal entries
- ▶ Key properties: $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$, $\det(\mathbf{A}') = \det(\mathbf{A})$
- ▶ $\mathbf{A}$ invertible $\iff \det(\mathbf{A}) \neq 0$

Summary - Part 3: Determinant and Rank (2/2)

6. Rank:

- ▶ Maximum number of linearly independent columns (or rows)
- ▶ $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$
- ▶ For square matrices: $\text{rank}(\mathbf{A}) = n \iff \det(\mathbf{A}) \neq 0$